



Assessment Analysis Guide
25-26 Math G3 Form I Mult and Div
Mathematics
Grade 3

Reference #: 682e2d8

1

Item Id: i184997

Item Type: SR (Selected Response)

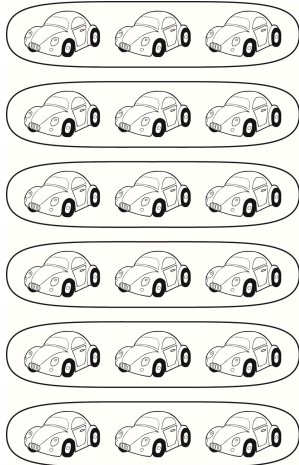
Standards Description:

3.OA.A.2

Interpret whole-number quotients of whole numbers, e.g., interpret $56 \div 8$ as the number of objects in each share when 56 objects are partitioned equally into 8 shares, or as a number of shares when 56 objects are partitioned into equal shares of 8 objects each.

Content:

Which equation describes how the toy cars are divided?



A $3 \div 6 = 18$

B $6 \div 3 = 18$

C $6 \div 18 = 3$

D $18 \div 6 = 3$

Primary Standard: 3.OA.A.2**Points Possible: 1****Distractor Rationale:**

A: $3 \div 6 = 18$

Student used a division symbol as if it were a multiplication symbol.

B: $6 \div 3 = 18$

Student used a division symbol as if it were a multiplication symbol.

C: $6 \div 18 = 3$

Student confused the dividend and divisor when representing an equation.

D: $18 \div 6 = 3$

Correct.

2

Item Id: i184965

Item Type: SR (Selected Response)

Standards Description:

3.OA.A.1

Interpret products of whole numbers, e.g., interpret 5×7 as the total number of objects in 5 groups of 7 objects each.

Content:

Felipe wrote this number sentence to solve a problem:

$8 \times 4 = ?$

Which problem was Felipe **most likely** trying to solve?

- A
- Jimena has 8 bracelets. She buys 4 more. How many bracelets does Jimena have?
- B
- Jimena has 8 muffins. She separates the muffins into 4 groups. How many muffins are in each group?
- C
- Jimena has 8 bags of grapes. There are 4 grapes in each bag. How many grapes does Jimena have?
- D
- Jimena has 8 stickers. Her friend has 4 stickers. How many more stickers does Jimena have than her friend?

Primary Standard: 3.OA.A.1

Points Possible: 1

Distractor Rationale:

- A:
- Jimena has 8 bracelets. She buys 4 more. How many bracelets does Jimena have?
- Student identified a context illustrating addition as one that would illustrate multiplication.
- B:
- Jimena has 8 muffins. She separates the muffins into 4 groups. How many muffins are in each group?
- Student identified a context illustrating division as one that would illustrate multiplication.
- C:
- Jimena has 8 bags of grapes. There are 4 grapes in each bag. How many grapes does Jimena have?
- Correct.
- D:
- Jimena has 8 stickers. Her friend has 4 stickers. How many more stickers does Jimena have than her friend?
- Student identified a context illustrating subtraction as one that would illustrate multiplication.

3

Item Id: i184998

Item Type: SR (Selected Response)

Standards Description:

3.OA.B.5 Apply properties of operations as strategies to multiply and divide.

Content:What is another way of expressing 3×9 ?

- A $(5 \times 4) + 3$
- B $(3 \times 5) + 4$
- C $(3 \times 5) + (3 \times 4)$
- D $(3 + 5) \times (3 + 4)$

Primary Standard: 3.OA.B.5**Points Possible:** 1**Distractor Rationale:**A: $(5 \times 4) + 3$

Student decomposed the multiplier into the right values but related those values with multiplication and then added the multiplicand.

B: $(3 \times 5) + 4$

Student correctly decomposed the multiplier, but only distributed the multiplicand to one of the resulting terms.

C: $(3 \times 5) + (3 \times 4)$ **Correct.**D: $(3 + 5) \times (3 + 4)$

Student attempted to rewrite a multiplication expression using the distributive property, but mistakenly switched the addition and multiplication symbols.

4

Item Id: i184996

Item Type: MS (Multiple Select)

Standards Description:

3.OA.A.2

Interpret whole-number quotients of whole numbers, e.g., interpret $56 \div 8$ as the number of objects in each share when 56 objects are partitioned equally into 8 shares, or as a number of shares when 56 objects are partitioned into equal shares of 8 objects each.

Content:

Select **two** statements that can be represented by the expression $15 \div 3$.

- A

Duncan buys 15 pencils. He gives away 3 pencils.
- B

Jaleel receives 15 stickers from each of 3 of his classmates.
- C

Natasha has 15 colored tiles. She places them in 3 equal rows.
- D

Kevin has 15 students in his class. The class gains 3 new students.
- E

Olivia gives a total of 15 muffins away. She gives the same number of muffins to each of 3 friends.

Primary Standard: 3.OA.A.2

Points Possible: 1

Distractor Rationale:

- A:

Duncan buys 15 pencils. He gives away 3 pencils.

Student identified a situation requiring subtraction as one requiring division.
- B:

Jaleel receives 15 stickers from each of 3 of his classmates.

Student identified a situation requiring multiplication as one requiring division.
- C:

Natasha has 15 colored tiles. She places them in 3 equal rows.

Correct.
- D:

Kevin has 15 students in his class. The class gains 3 new students.

Student identified a situation requiring addition as one requiring division.
- E:

Olivia gives a total of 15 muffins away. She gives the same number of muffins to each of 3 friends.

Correct.

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Item Id: i184964

Item Type: SR (Selected Response)

Standards Description:

3.OA.A.1 Interpret products of whole numbers, e.g., interpret 5×7 as the total number of objects in 5 groups of 7 objects each.

Content:

Valentina makes a painting that has 5 rows of stars and 9 stars in each row, shown below.



Which expression represents the number of stars on Valentina's painting?

A $5 + 5 + 5$

B 5×9

C $9 + 5$

D $9 \div 5$

Primary Standard: 3.OA.A.1**Points Possible:** 1**Distractor Rationale:**

A: $5 + 5 + 5$

Student may have thought that only repeated addition, and not multiplication, can represent arrays; chose an expression in which a factor is repeatedly added an incorrect number of times.

B: 5×9

Correct.

C: $9 + 5$

Student may not have understood how an array relates to multiplication; chose an expression in which the factors

D: $9 \div 5$

are added rather than multiplied.

Student confused a context calling for multiplication with one calling for division; divided instead of multiplying the factors.

6

Item Id: i185000

Item Type: SR (Selected Response)

Standards Description:

3.OA.B.5 Apply properties of operations as strategies to multiply and divide.

Content:In the expressions below, d represents an unknown number.Which expression is equivalent to $5 \times d$?

A $d + 5$

B $d \times 5$

C $5 \div d$

D $d \div 5$

Primary Standard: 3.OA.B.5**Points Possible:** 1**Distractor Rationale:**

A: $d + 5$

Student misunderstood the relationship between multiplication and addition; thought that multiplying two factors is equivalent to adding them in the reverse order.

B: $d \times 5$

Correct.

C: $5 \div d$

Student misunderstood the relationship between multiplication and division; may have thought they were equivalent operations.

D: $d \div 5$

Student misunderstood the relationship between multiplication and division; chose an expression that represents the product $a \times b$ as equal to the quotient b / a .